4.5 SPEARMAN'S RANK CORRELATION COEFFICIENT

If the data are in ordinal scale then Spearman's rank correlation coefficient is used. It is denoted by the Greek letter ρ (**rho**).

Spearman's correlation can be calculated for the subjectivity data also, like competition scores. The data can be ranked from low to high or high to low by assigning ranks.

Spearman's rank correlation coefficient is given by the formula

$$\rho = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n\left(n^2 - 1\right)}$$

where $D_i = R_{1i} - R_{2i}$

R_{1i} = rank of i in the first set of data

R_{2i} = rank of i in the second set of data and

n = number of pairs of observations

Interpretation

Spearman's rank correlation coefficient is a statistical measure of the strength of a monotonic (increasing/decreasing) relationship between paired data. Its interpretation is similar to that of Pearson's. That is, the closer to the ±1 means the stronger the monotonic relationship.

Positive Range	Negative Range
0.01 to 0.19: "Very Weak Agreement"	(-0.01) to (-0.19): "Very Weak Disagreement"
0.20 to 0.39:"Weak Agreement"	(-0.20) to (-0.39): "Weak Disagreement"
0.40 to 0.59: "Moderate Agreement"	(-0.40) to (-0.59): "Moderate Disagreement"
0.60 to 0.79: "Strong Agreement"	(-0.60) to (-0.79): "Strong Disagreement"
0.80 to 1.0: "Very Strong Agreement"	(-0.80) to (-1.0): "Very Strong Disagreement"

Example 4.3

Two referees in a flower beauty competition rank the 10 types of flowers as follows:

Referee A	1	6	5	10	3	2	4	9	7	8
Referee B	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient and find out what degree of agreement is between the referees.

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